

Lab-6: 4-bit Common Bus System

Aim:

To design and implement a 4-bit common bus system using multiplexers and registers.

Theory:

A **Common Bus** is a shared communication pathway that allows multiple registers to transfer data to one another. Only one register can place data on the bus at a time, controlled by selection logic.

In this design:

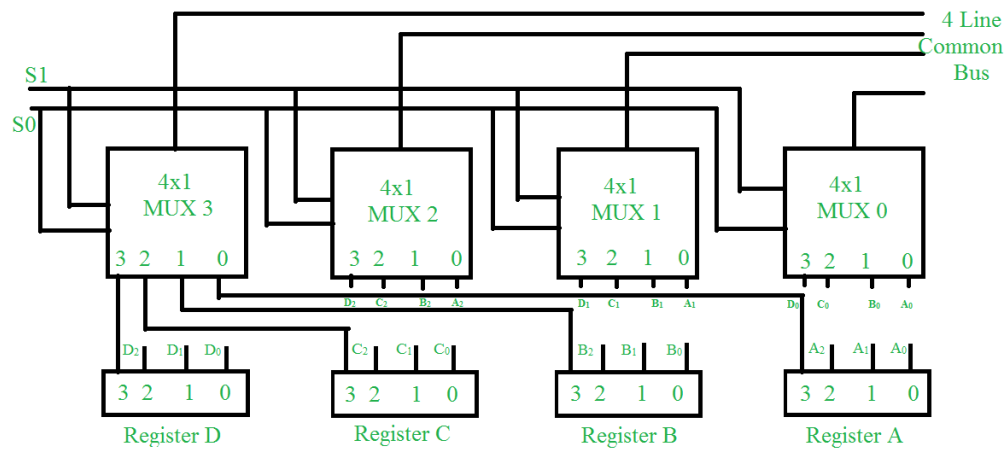
- **Registers (D Flip-Flops with WE)** store 4-bit data. The WE (Write Enable) signal controls when a register loads data from the bus. The R (Reset) signal clears the register.
- **Multiplexers** act as bus selectors — they determine which register's output is placed onto the common bus at any given time. The select lines of all MUXes are driven by a common control input.
- **4 Registers (R0–R3)** are connected — each can be a source (via MUX) or destination (via WE) of data on the bus.

Data Transfer Operation:

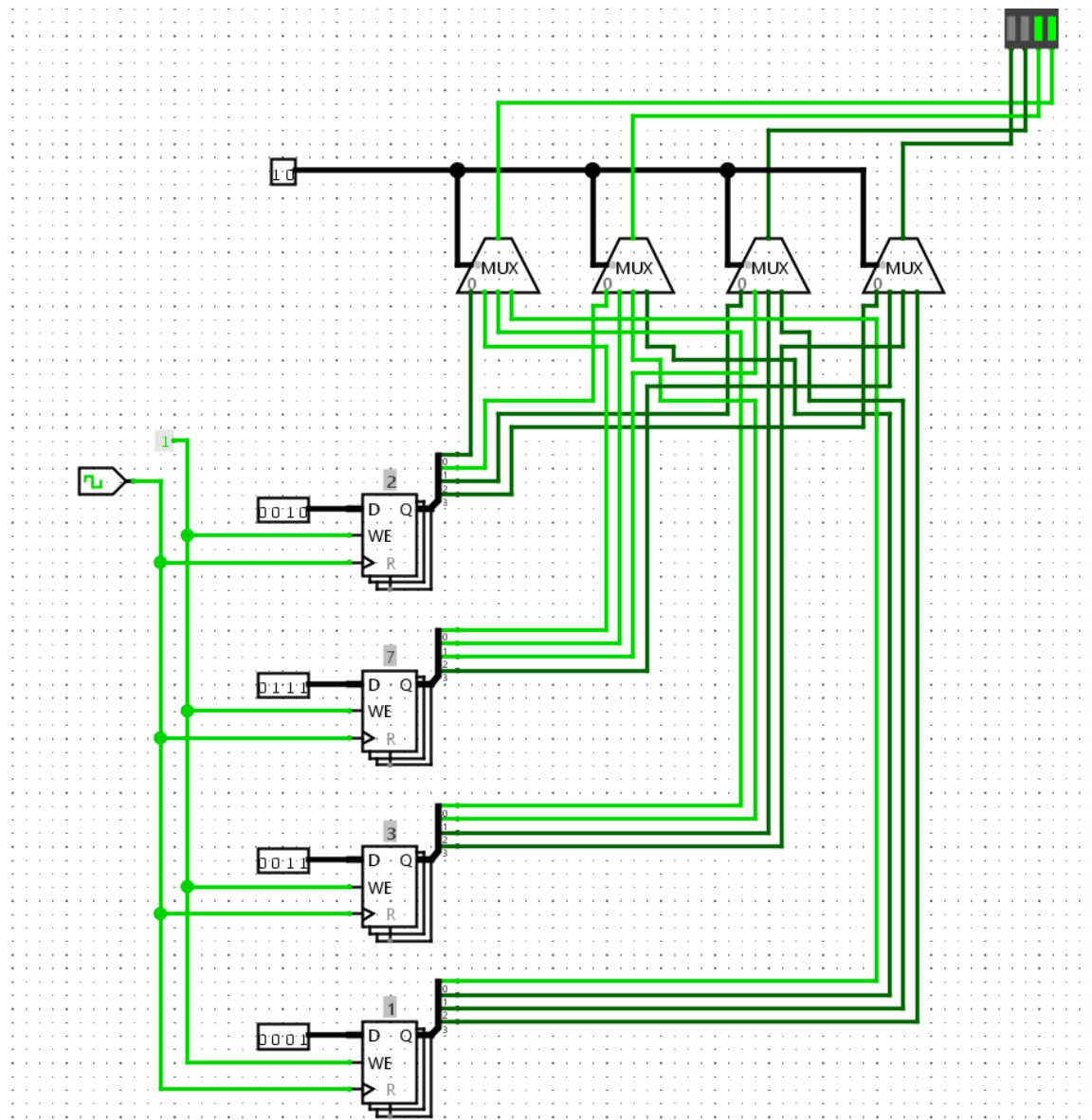
- To transfer data from Register Rx to Register Ry:
 - Set MUX select lines to point to Rx
 - Assert WE on Ry
 - On the clock edge, Ry loads the value present on the bus (i.e., Rx's output)

Procedure:

1. Open Logisim and create a new project.
2. Place four 4-bit registers (D flip-flops with WE and Reset inputs), storing values 6 (0110), 15 (1111), 7 (0111), and 0 (0000).
3. Connect the Q output of each register to the corresponding input of four MUX components.
4. Connect all MUX select lines to a common 2-bit control input.
5. Wire the MUX outputs to form the common bus.
6. Connect the bus output back to the D input of all registers.
7. Use WE signals to select the destination register.
8. Apply clock pulses and observe data transfer via the output display.



Circuit Diagram:



Conclusion:

The 4-bit Common Bus system was successfully designed and simulated in Logisim. The MUX select lines correctly routed the output of the selected register onto the shared bus, and the destination register loaded the bus data on the clock edge when WE was asserted. This experiment demonstrates the fundamental concept of bus-based data transfer used in CPU register files and datapath design.